

ASSIGNMENT

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

•**Native** – Apps are made for only one operating system, like Android or iOS. They are fast and have the best performance. It is commonly used for games or apps that need high speed and device features.

•**Hybrid** – Apps are built using web technologies and can run on different platforms. It is commonly used for simple apps that need to be developed quickly.

•**Cross-platform** – Apps are built once and can run on both Android and iOS using frameworks like Flutter. It is commonly used when developers want to save time and reduce development costs.

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

- The **Flutter Ecosystem** is the collection of tools, libraries, packages, and services used in Flutter development. It helps developers build attractive, fast, and responsive mobile apps with one codebase that works on both Android and iOS.

3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?

- The **Software Development Life Cycle (SDLC)** is a step-by-step process for creating software. It includes planning, designing, developing, testing, deploying, and maintaining the application. It is important because it keeps the project organized, reduces errors, and helps create a high-quality application.

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

- Flutter provides the tools to build the application, while the SDLC provides the process to develop it properly. Together, they help developers create apps that are organized, tested, and ready for users.

5. Why should developers understand these concepts before starting an application project?

- Developers should understand these concepts because they help them choose the right development approach, use the right tools, and follow a proper process. This makes it easier to build a successful, efficient, and user-friendly application.